RoadFury: Transformer-Based Test Prioritization for Simulation-Based Testing of Self-Driving Cars

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***Abstract*—We present RoadFury, a test prioritization tool for simulation-based regression testing of self-driving cars. In BeamNG.tech simulations, executing large test suites is prohibitively expensive; prioritization reorders tests so that failure-revealing cases execute first. RoadFury extracts a 10-channel sequential feature matrix from road geometry—curvature, heading sin/cos, curvature jerk, curvature acceleration, local variability, and positional encodings—and processes it with a Pre-LN Transformer Encoder with learnable `[CLS]` token, trained with Stochastic Weight Averaging (SWA) for robust generalization. Deployed as a Docker-containerized gRPC service for the ICST 2026 SDC Testing Competition [7], RoadFury achieves APFD = 0.8042 ± 0.0120, surpassing ITEP4SDC (0.7812) and random ordering (0.4930).**

***Index Terms*—test prioritization, self-driving cars, Transformer, stochastic weight averaging, road geometry**

I. Introduction

Simulation-based testing of SDCs in high-fidelity simulators (e.g., BeamNG.tech) is indispensable for validation yet expensive [1], [2]: individual tests run for tens of seconds to minutes. The rapid ascent of AI agents in autonomous navigation [11] widens this gap further. *Test prioritization* addresses this by finding a permutation π that maximizes early fault detection, measured by APFD [3]:

$$\text{APFD}(\pi) = 1 - \frac{\sum_{i=1}^{m} TF_i}{n \cdot m} + \frac{1}{2n} \tag{1}$$

where n is total tests, m failing tests, TF_i the rank of the i-th fault (1.0: all failures first; 0.5: random).

Prior tools—SDC-Scissor [2] and ITEP4SDC—use 3 hand-crafted features (segment length, angle change, curvature) in feedforward networks, discarding the *sequential ordering* of road geometry. **RoadFury** addresses this gap with: (1) a *10-channel geometric feature representation* encoding higher-order curvature dynamics and positional context; and (2) a *Transformer Encoder with SWA* capturing long-range inter-point dependencies. A systematic evaluation [10] confirms Transformers consistently outperform classical classifiers on sequential feature inputs; analogous early-stage ML prediction of AI-artifact outcomes is validated in related work [9]. SDC-Scissor [1], [2] pioneered road-feature ML for test selection; Yoo and Harman [3] survey regression testing strategies; Izmailov et al. [5] show SWA finds wider optima; broader contextual enrichment benefits are shown in [12].

TABLE I
10-Channel Road Geometry Features

Ch.	Feature	Description
f_0	Segment length	$\lVert p_{i+1} - p_i \rVert_2$
f_1	Abs. angle change	$\lvert \Delta\theta_i \rvert$
f_2	Menger curvature	$\kappa_i = 4A_i/(a_i b_i c_i)$, Heron's formula
f_3	Curvature jerk	$\Delta\kappa_i$, 1st derivative of κ
f_4	Cum. dist. norm	$\sum_{j<i} \lVert p_{j+1} - p_j \rVert / d_{\text{tot}} \in [0,1]$
f_5	Heading sin	$\sin(\theta_i)$, $\theta_i = \text{atan2}(\Delta y, \Delta x)$
f_6	Heading cos	$\cos(\theta_i)$, avoids $\pm\pi$ wrap
f_7	Rel. position	$i/(L-1) \in [0,1]$
f_8	Local curv. std	$\text{std}(\kappa_{i-5:i+5})$, window = 11
f_9	Curv. accel.	$\Delta^2\kappa_i$, 2nd derivative of κ

II. Approach

A. Road Normalization and Feature Extraction

Each test case is a variable-length sequence of 2D road control points. We resample all sequences to L=197 points via linear interpolation and extract 10 intrinsic geometric channels (Table I), yielding a (197×10) translation- and rotation-invariant feature matrix, z-normalized with training-set statistics.

Channels f_3 (jerk) and f_9 (acceleration) encode curvature transition dynamics; f_8 detects oscillatory segments; f_5, f_6 provide unambiguous heading. This contextual enrichment via higher-order representations improves prediction accuracy [12].

B. RoadTransformer Architecture

The (197×10) matrix passes through (Fig. 1):

(1) Input projection: $\mathbf{h}_i = \text{GELU}(\text{LN}(\mathbf{W}_p\mathbf{f}_i + \mathbf{b}_p))$, projecting 10-d to d_{m}=128.

(2) CLS + positional embedding: Learnable `[CLS]` $\in \mathbb{R}^{128}$ prepended (len. 198); positional embeddings $\mathbf{E} \in \mathbb{R}^{198\times128}$ added.

(3) Pre-LN Transformer Encoder [8]: 4 layers, 8 heads, FFN 512, GELU, dropout 0.1:

$$\begin{aligned} \mathbf{Z}' &= \mathbf{Z}^{(l)} + \text{MHA}(\text{LN}(\mathbf{Z}^{(l)})), \\ \mathbf{Z}^{(l+1)} &= \mathbf{Z}' + \text{FFN}(\text{LN}(\mathbf{Z}')) \end{aligned} \tag{2}$$

Self-attention relates all 197 positions globally.

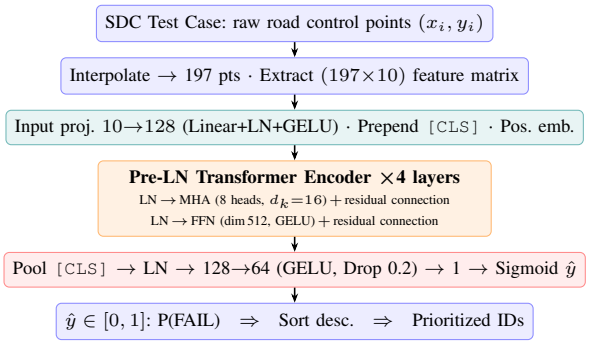

Fig. 1. RoadFury architecture. Road geometry is resampled to 197 points and encoded as a (197×10) feature matrix, then processed by a four-layer Pre-LN Transformer Encoder with a learnable `[CLS]` token. The `[CLS]` output is classified by an MLP head yielding failure probability $\hat{y}$, used to sort tests descending.

TABLE II
TEST PRIORITIZATION PERFORMANCE

Tool	APFD	#Feats	Architecture
Random	0.493	–	–
ITEP4SDC	0.781	3	ONNX MLP
RoadFury	$\mathbf{0.804 \pm 0.012}$	10	Transformer+SWA

(4) MLP classifier: CLS output → LN → Linear(128→64) → GELU → Drop(0.2) → Linear(64→1) → Sigmoid. Output $\hat{y}\in[0, 1]$ is failure probability (∼400K params total).

C. Training (SWA) and Deployment

Adam (lr = 1e-3, BCE), 75 epochs on 28,804 samples. SWA [5] starts at epoch 50—averaging weights over the final 25 epochs finds a flatter loss basin, improving robustness to unseen road geometries. The SWA checkpoint (`roadfury_best.pt`) stores the averaged state dict and per-channel normalization statistics.

At inference, n test cases are batched $(n, 10, 197)$, scored in one forward pass, sorted by descending $\hat{y}$, and streamed back via gRPC. The service implements `Name()`, `Initialize(stream Oracle)` (acknowledged; no online retraining), and `Prioritize(stream SDCTestCase)`. Launched via: `docker run --rm -p 4545:4545 road-fury -p 4545`

III. EVALUATION

Dataset. SensoDat [6]: 36,006 labeled BeamNG simulation tests (RF = 1.5, max speed 120 km/h, OOB = 50%), split 80/20 (28,804 train / 7,202 val).

Baselines. (1) **Random**: uniform shuffle, APFD ≈ 0.5. (2) **ITEP4SDC**: prior competition SoA—ONNX feedforward MLP, 3 features, pre-fitted scalers.

RoadFury achieves APFD = 0.804 (σ = 0.012), outperforming ITEP4SDC by **+2.9%** and random by **+63%**. Low σ across trials confirms SWA's flat-basin stability.

Key factors: (1) *Richer feature dynamics*—f_3 (jerk) and f_9 (acceleration) capture the onset and rate of curvature change at each road point, detecting dangerous transitions invisible to ITEP4SDC's 3 static features. f_8 identifies oscillatory segments where autopilots repeatedly over-steer. (2) *Sequential attention*—ITEP4SDC processes points independently; RoadFury relates all 197 positions simultaneously, recognizing compound patterns (long straight accelerating into a hairpin) whose danger exists only in their combination. This global structural awareness directly translates to earlier fault detection and higher APFD.

IV. CONCLUSION

RoadFury achieves APFD = 0.804 ± 0.012 on the ICST 2026 SDC Competition benchmark—a new state of the art over the ONNX-based ITEP4SDC baseline. By uniting a 10-channel road geometry representation, a Pre-LN Transformer Encoder, and Stochastic Weight Averaging, it captures both local curvature dynamics and global sequential structure that feedforward baselines discard. Future work includes online oracle adaptation (few-shot fine-tuning on `Initialize` oracle data) and feature-subset ablation. Tool available at: https://github.com/chisngyen/sdc-test-prioritization-novel.

REFERENCES

[1] C. Birchler, S. Khatiri, B. Bosshard, A. Gambi, and S. Panichella, "Machine learning-based test selection for simulation-based testing of self-driving cars," *EMSE*, vol. 28, no. 71, 2023.
[2] C. Birchler, N. Ganz, S. Khatiri, A. Gambi, and S. Panichella, "Cost-effective simulation-based test selection with SDC-Scissor," in *Proc. IEEE SANER*, 2022, pp. 386–390.
[3] S. Yoo and M. Harman, "Regression testing minimization, selection and prioritization: A survey," *STVR*, vol. 22, no. 2, pp. 67–120, 2012.
[4] A. Vaswani et al., "Attention is all you need," in *NeurIPS*, vol. 30, 2017.
[5] P. Izmailov, D. Podoprikhin, T. Garipov, D. Vetrov, and A. G. Wilson, "Averaging weights leads to wider optima and better generalization," in *Proc. UAI*, 2018.
[6] C. Birchler, S. Khatiri, and S. Panichella, "SensoDat: Simulation-based sensor dataset of self-driving cars," 2024.
[7] P. Aryan, T. Fulcini, L. Starace, C. Birchler, and S. Panichella, "ICST Tool Competition 2026 — SDC Testing Track," in *Proc. IEEE ICST*, 2026.
[8] R. Xiong et al., "On layer normalization in the Transformer architecture," in *Proc. ICML*, 2020.
[9] D. S. Duy Minh, T. K. Huynh, et al., "Early-stage prediction of review effort in AI-generated pull requests," in *Proc. IEEE/ACM MSR*, 2026.
[10] T. K. Huynh, D. S. Duy Minh, et al., "Systematic evaluation of ML and Transformer methods for scientific literature classification," in *Proc. TRACS@WASP*, 2025.
[11] P. H. Pham, C. N. Tran, D. S. Duy Minh, et al., "Navigating simply, aligning deeply: Winning solution for Mouse vs. AI 2025," in *NeurIPS 2025 Competition Track*, 2026.
[12] L. P. Q. Nguyen, D. S. Duy Minh, et al., "Beyond vision: Contextually enriched image captioning," in *ACM MM EVENTA*, 2025. arXiv:2512.20042.